Weather Monitoring Portal

Test Plan — Sydney Metro M1

Every feature promised in the MTS proposal, and the tests that prove it: the portal, the alert engine, pump control, history, users and security.

Against proposal OBS-MTS-M1-WX-2026-01

29 October 2026

This test plan lists every feature promised in the **MTS proposal** for the Sydney Metro M1 weather monitoring portal and server (OBS-MTS-M1-WX-2026-01), with the tests that prove each one.

Each test case has a **Result** column for the tester to record Pass, Fail or Blocked.

Contents

1 About this plan

2 How testing works

3 Feature list

4 Test cases — map, trends and flood event

5 Test cases — station page and pumps

6 Test cases — alerts and rules

7 Test cases — historical query and CSV export

8 Test cases — users, roles, sign-in and audit

9 Test cases — system health, data and fallback

10 Test cases — responsive layout and system qualities

11 Sign-off

1. About this plan

This plan lists every feature the MTS proposal promises for the weather monitoring portal and server, and how each one will be tested. It is the checklist both sides use to agree that the system does what was proposed.

- **Tested against:** *Weather Monitoring — Sydney Metro M1*, proposal OBS-MTS-M1-WX-2026-01 (22 July 2026). Section numbers in this plan refer to it.
- **In scope:** the online portal, the central server and its alert engine, and the software behaviour of the stations as the portal sees it (pump control, fallback, store-and-forward).
- **Out of scope here:** sensor accuracy, mast and enclosure build, solar sizing and EMC. These are proven by calibration certificates and the per-location site acceptance test (proposal §10.1, stage 6). The feature list still names them, so nothing is missed.
- **Requirements still to come:** MTS's detailed requirements will refine thresholds, wording and recipients. Test values marked *(pending)* are the proposal's example values and change when MTS confirms them.

2. How testing works

Each test has an ID, the steps, the expected result, where it can run, and a result to record.

Where a test runs

Where	Meaning
Prototype	Can be checked now on the design prototype, which runs on demonstration data. Proves the screen and the user journey, not the live data behind it
System	Needs the built server with real or simulated station data: alert timing and delivery, pump commands, ingest, fallback, security
Site	Needs the installed stations, during commissioning and the site acceptance test

Test data. Server tests use a station simulator that replays rainfall, level, temperature and wind series, so each threshold can be crossed on demand. Site tests use the real sensors, the manual staff gauge and the existing Marrickville pumps.

Pass or fail. A test passes when every point of its expected result is seen. Anything else is a fail, recorded as a defect with steps to reproduce and a screenshot.

Defect severity

Severity	Meaning	Release
Critical	A safety function is wrong or missing: an alert not raised, wrong wording, a pump command not supervised	Blocks release
Major	A feature does not work, with no workaround	Blocks release
Minor	Works with a workaround, or a display fault	Fixed before handover
Cosmetic	Look and wording only	Fixed when agreed

3. Feature list

The proposal promises 30 features in scope, and MTS has asked for 7 more (9 Nov 2026, marked *MTS request*). Each is covered by the tests named in the last column.

#	Feature	Proposal	Where	Tests
1	Corridor map: every station's readings against its alert thresholds	§8.1, Fig 6	Prototype, System	MAP-01 to 04
2	Station-status table: readings, pump, battery, signal, last update, status	Fig 6	Prototype, System	MAP-05, 06
3	Live header: Sydney time, active-alert count, stations online, alert banner	Fig 6	Prototype, System	MAP-07, 08
4	Trends: wind mean and gust, rainfall, temperature, humidity, with dashed thresholds	Fig 7	Prototype, System	TRD-01 to 04
5	Flood-event screen: two-location level series, pumps, vicinity inset, event log	Fig 8	Prototype, System	FLD-01 to 03
6	Station page: status summary, readings grouped by theme with charts, station health, level trend against thresholds	§5.6, Fig 12	Prototype, System	STN-01 to 03
7	Pump status from the pump panel's run signal: ON in green, OFF, since when, today's run time	§5.5, MTS request	Prototype, System	PMP-01, 04, 05
8	Pump run log: start and stop time, water level at start and at stop, last 7 days	MTS request	Prototype, System	PMP-02, 03, 07
9	Pumps monitored only: no control in the portal; run signal wired from the existing panel	MTS request	Prototype, Site	PMP-06, 08
10	Rainfall vigilance: 25 mm/1 h, 45 mm/3 h, 120 mm/3 days, countdown, reset, all-clear	§7.2	System	ALR-01 to 05
11	Flood, temperature and wind alerts with the required wording	§7.3	System	ALR-06 to 09
12	Alert delivery by screen, email and SMS: weather within 5 min, faults within 30 min, required content	§7.1	System	ALR-10 to 13
13	Alert and notification history: filter, acknowledge, badge, CSV export	§7.6, Fig 15	Prototype, System	ALR-14 to 18
14	Annotate alerts	§8.1	Prototype, System	ALR-19
15	Rules engine: edit, add, enable, disable, schedule; validated, versioned and audited	§7.3, Fig 17	Prototype, System	RUL-01 to 07
16	Redundancy and fallback: alternate source, reworded coverage, fault alert	§7.4	System	FBK-01 to 04
17	Historical query by location, parameter, dates and interval; paged, sortable; CSV	§8.3, Fig 10	Prototype, System	HIS-01 to 05
18	User management: add with invitation, edit, suspend, remove, station scope	§8.4, Fig 14	Prototype, System	USR-01 to 05
19	Roles: six standard roles, several per user, custom roles, least privilege	§8.4	System	USR-06 to 08

#	Feature	Proposal	Where	Tests
20	Sign-in security: password policy, lockout, inactivity timeout, optional MFA, re-check on every action, encrypted sessions	§8.4	System	USR-09 to 14
21	20 or more concurrent named logins	§8.1, §8.4	System	NFR-01
22	Audit trail: users, roles, thresholds, acknowledgements, recipient groups; queryable, exportable	§8.4	System	AUD-01 to 03
23	System health: inoperable, unresponsive, missing sensor or pump data, low battery; refreshed ≤ 10 min; maintenance mode	§8.1, §9	Prototype, System	HLT-01 to 04, 07
24	Power telemetry: state of charge, PV input, charge/discharge, fault flags, alarmed	§8.6	System, Site	HLT-05, 06
25	Data ingest: timestamping, event-triggered messages, store-and-forward with no data loss	§8.2a, §8.5	System	ACQ-01 to 03
26	Data validation: range and rate limits, quarantine, data-quality alert, radar–float cross-check	§8.2b	System	ACQ-04 to 06
27	Processing: 10-min, hourly and multi-day totals	§8.2c	System	ACQ-07
28	Responsive portal: phone, tablet, laptop; pump screen on a phone	§8.1, Fig 9, Fig 13	Prototype, System	RSP-01 to 04
29	Availability, 48 h maintenance notice, cloud hosting with backups	§8.1, §9	System	NFR-02 to 04
30	Permanent data retention	§8.3	System	NFR-05
31	Station menu: a list of every station to go straight to	MTS request	Prototype, System	MAP-09
32	Map backgrounds: plain schematic, street map or satellite (Mapbox)	MTS request	Prototype, System	MAP-10, 11
33	Station quick view over the map	MTS request	Prototype, System	MAP-12
34	Map legend and station cards that minimise and expand	MTS request	Prototype, System	MAP-13
35	Battery and signal shown on demand, not by default	MTS request	Prototype, System	MAP-14
36	Station health: last connected time; "no data" after 15 minutes	MTS request	Prototype, System	STN-04, 05
37	Recipient groups: add, edit members and channels, remove	§7.3, MTS request	Prototype, System	RCP-01 to 04

Proven at site acceptance, not in this plan: sensor accuracy and ranges (§5.1–5.4), the manual staff gauge (§5.2), mast, enclosure and KE+200 clearance (§5.5, §5.7), solar autonomy (§8.6), EMC (§8.5), calibration certificates (§11.1).

Removed by MTS (9 Nov 2026): pump control (§5.6 local control on the OMC-048, §8.4 supervised manual start/stop). The existing pump panel runs the pumps; the portal receives each pump's run signal and monitors only.

Removed by MTS (5 Nov 2026): the PTZ camera verification link (§7.5). There is no integration between the system and the cameras, so the portal has no camera pop-up, icon or feed link. The standing-water alert remains (ALR-06).

To confirm with MTS: the proposal lists SMS as an alert channel; the agreed development scope (Rev B) delivers screen, email and web push instead. Tests ALR-10 to 13 assume the confirmed channels. The

proposal also mentions sending alerts to other connected systems, such as operational displays or third-party interfaces (§8.2d); MTS to name any it needs, and a test is added for each.

4. Test cases — map, trends and flood event

Corridor map and station status (MAP)

ID	Test	Steps	Expected result	Where	Result
MAP-01	**Every station on the map**	Open **Map**	All seven locations shown in corridor order, each with a numbered pin and a reading card	Prototype, System	
MAP-02	**Readings against thresholds**	Compare each card with the station's latest readings	Each card shows only the sensors fitted there (§4.3), as value / threshold with units; Lady Game Drive shows both its up and down tunnel readings	Prototype, System	
MAP-03	**Status colours**	Drive one station to warning, one to alert, stop one logger	Pins and cards turn amber, red and grey; each colour also has a word or icon	System	
MAP-04	**Map refresh**	Change a reading at the simulator	The map shows the new value within the refresh interval, without reloading the page	System	
MAP-05	**Status table contents**	Read the table under the map	One row per location: rain, wind, temperature, level, pump, updated, status	Prototype, System	
MAP-06	**Missing values**	Check a station without a sensor	Shows "–", never 0	Prototype, System	
MAP-07	**Header**	Read the top bar	Sydney time with AEST/AEDT, active-alert count, stations online	Prototype, System	
MAP-08	**Alert banner**	Raise an alert	The banner under the header shows the alert's instruction; **Open** goes to it	Prototype, System	
MAP-09	**Station menu**	Open **Station** in the top menu; pick Canterbury. On a phone, open the menu	All seven stations listed; it opens Canterbury; the current station is highlighted; the phone menu lists them too	Prototype, System	
MAP-10	**Map backgrounds**	Choose **Plain**, **Map**, then **Satellite**; reload	Schematic, street map and satellite each show all seven stations; the choice is remembered	Prototype, System	
MAP-11	**Pins on the map**	On Map, hover a pin, then click it	Pins carry the station number in its status colour; hover shows the live readings; click opens the quick view	Prototype, System	
MAP-12	**Station quick view**	Click a station on any map view; select **Open station page**; Ctrl-click a station	A panel over the map: readings, pumps, last connected, latest alerts; the button opens the station; Ctrl-click opens it in a new tab	Prototype, System	
MAP-13	**Legend and cards**	Minimise the legend; close one card with its **×**, then click its pin; click **Hide all cards**; reload; switch view; show all	The legend shrinks to a "Legend" tab; the × closes only that card and its pin brings it back; Hide all cards leaves only the numbered pins; the choices stay after reload and across Plain, Map and Satellite	Prototype, System	
MAP-14	**Battery and signal on demand**	Open the map; select **Show station health**; reload	Hidden at first; the button adds Battery and Signal to the table and the map cards; the choice is remembered	Prototype, System	

Trends (TRD)

ID	Test	Steps	Expected result	Where	Result
TRD-01	**Four charts**	Open **Trends**, pick Belmore triangle	Wind (mean solid, gust dashed), rainfall, temperature and humidity, each with a "now" value	Prototype, System	
TRD-02	**Thresholds drawn**	Read each chart	Alert thresholds dashed and labelled (75 km/h, 25 mm/hr, 38 °C) *(pending)*; rainfall bars over the threshold in red	Prototype, System	
TRD-03	**Station, range and interval**	Change each control	Charts redraw for the new station, period and averaging interval	Prototype, System	
TRD-04	**Export**	Select **Export CSV**	A CSV of the plotted data, with units	Prototype, System	

Flood event (FLD)

ID	Test	Steps	Expected result	Where	Result
FLD-01	**Two-location level series**	Raise water at Marrickville and Marrickville–Dulwich Hill	Both series on one axis with pump-start, high-high and rail-foot lines	Prototype, System	
FLD-02	**Pumps and vicinity**	During the event	Duty then standby shown running; the vicinity inset marks the affected location	Prototype, System	
FLD-03	**Event log**	Read the log	Every step timestamped in order: rate of rise, standing water, pump starts, block line	Prototype, System	

5. Test cases — station page and pumps

Station page (STN)

ID	Test	Steps	Expected result	Where	Result
STN-01	**Status summary**	Open **Station** for Marrickville, then for Campsie	Marrickville: a red line at the top lists rainfall, water level and float switch as over threshold, each linking to its panel. Campsie: "All readings within their thresholds"	Prototype, System	
STN-02	**Readings grouped by theme**	Open each of the 7 stations	Only the panels for the sensors fitted, always in the order Flood, Rain, Wind, Weather (Marrickville adds Pumps). Each shows its main value, status and a 3 h or 24 h chart with the threshold drawn; a panel over threshold has a red edge; rain shows the 1 h / 3 h / 3 day rules as bars; sensor model is behind the ⓘ	Prototype, System	
STN-03	**Level trend**	Read the 3-hour chart in the Flood panel	Level against pump-start (+100 mm) and high-high (+180 mm) *(pending)*, pump starts marked	Prototype, System	

ID	Test	Steps	Expected result	Where	Result
STN-04	**Station health and last connected**	Read the **Station health** line under the summary on each station	Battery and charging, cellular signal and **Last connected** date and time for each logger; no enclosure line	Prototype, System	
STN-05	**No data**	Stop a logger's telemetry for over 15 minutes	The health line shows "No data since [time] — logger or network", the summary names the logger, and the station's badge changes to **No data**	System	

Pump monitoring (PMP)

ID	Test	Steps	Expected result	Where	Result
PMP-01	**Pump ON and OFF**	Switch the duty pump's run signal ON, then OFF	ON shows the pump in green with the time it started; OFF shows grey with the time it last stopped	Prototype, System	
PMP-02	**Run log: start**	Switch a pump's run signal ON	A new row in the run log with the start time and the water level at that moment	Prototype, System	
PMP-03	**Run log: stop**	Switch it OFF again	The row gets the stop time, how long it ran and the water level at stop	Prototype, System	
PMP-04	**Both pumps**	Run duty and standby together	Both green; the card header says RUNNING; each has its own row in the log	Prototype, System	
PMP-05	**Today's totals**	Compare the card with the log	Today's run time and number of starts on each pump match its runs in the log	Prototype, System	
PMP-06	**No pump control**	Look for pump controls as every role, Administrator included	There is no AUTO/MANUAL switch and no Start or Stop button anywhere	Prototype, System	
PMP-07	**Run history**	Open the log after several runs	Last 7 days, newest first; **Show all** reveals older runs; weekly exercise runs are labelled	Prototype, System	
PMP-08	**Run signal from the panel**	Run each pump from the existing panel on site	The portal shows it ON within the reporting interval and OFF when it stops, with the matching log rows	Site	

6. Test cases — alerts and rules

Alert logic, delivery and history (ALR)

ID	Test	Steps	Expected result	Where	Result
ALR-01	**Rain 25 mm in 1 h**	Simulate 26 mm in 50 min at Marrickville	Alert raised at once with the CJC-T patrol wording (Hurlstone Park–Bankstown)	System	
ALR-02	**Rain 45 mm in 3 h and 120 mm in 3 days**	Simulate each total	Each raises its own alert; 45 mm adds the 60 kph TSR and demobilise-patrols reminder	System	

ID	Test	Steps	Expected result	Where	Result
ALR-03	**Vigilance countdown**	Let rain fall back below each threshold	Alert stays in force for 6 h, 12 h or 48 h from when rain fell below the line	System	
ALR-04	**Reset on re-trigger**	Add a qualifying burst inside the countdown	The full countdown restarts from the beginning	System	
ALR-05	**All-clear**	Let a countdown reach zero	A cancellation alert is sent and the whole sequence is in the event history	System	
ALR-06	**Flood: standing water**	Raise level above the standing-water set point	Immediate alert with the verification wording; no camera link	System	
ALR-07	**Flood: above rail foot and trending down**	Raise above rail foot, then let it fall	"Block the line" at once; on the fall, system checks, track inspection, staged 25 → 60 kph → unrestricted	System	
ALR-08	**Heat**	Hold 38 °C then 45 °C for over 5 min; fall back for over 10 min	60 / 40 kph TSR and heat-patrol wording on the rise; each stands down only after 10 min below	System	
ALR-09	**Wind**	Gusts 75, 85 and 120 kph	TSR wording at the listed kilometrages at 75 and 85; block-line wording on external track at 120	System	
ALR-10	**Weather alert within 5 min**	Time from breach to arrival on screen and email	Every channel within 5 minutes; breach transmitted immediately, not on the 5-minute cycle	System	
ALR-11	**Fault alert within 30 min**	Stop a logger	System-fault alert on every channel within 30 minutes	System	
ALR-12	**Alert content**	Read a delivered alert	Date and time identified, track and rail (up/down), station kilometrage, hyperlink that opens the event	System	
ALR-13	**Recipients and retries**	Make one recipient's email fail	Others receive it; the failed one is retried and the failure recorded	System	
ALR-14	**Alerts panel**	Open **Alerts**	Tiles for active, unacknowledged, today and last 7 days; event log newest first	Prototype, System	
ALR-15	**Filters**	Filter by severity, unacknowledged, location, date range	The list and counts match the filter	Prototype, System	
ALR-16	**Every event type**	Generate one of each	Rain, flood, pump, temperature, wind, sensor fault, data quality and power events all logged	System	
ALR-17	**Acknowledge**	Acknowledge one alert, then **Mark all**	Each shows who and when; header badge and counts drop at once	Prototype, System	
ALR-18	**Export**	**Export CSV**	The filtered events with acknowledgement details	Prototype, System	
ALR-19	**Annotate**	Add a note to an event	Note saved with name and time, shown on the event	Prototype, System	

The 3-day rainfall rule also has six detailed scenarios with test data (R3D-01 to 06: the portal's **Admin → Rain rule test**, and `mts/docs/rain-3day-tests`).

Rules engine (RUL)

ID	Test	Steps	Expected result	Where	Result
RUL-01	**Rules table**	Open **Admin → Alert rules**	Every rule with parameter, trigger, dwell/vigilance, severity, on/off	Prototype, System	
RUL-02	**Edit a rule**	Change a threshold, wording, locations, recipients and channels; save	The engine uses the new values at once, with no software change	Prototype, System	
RUL-03	**Validation**	Save an empty value or a warning above its alert	Refused with the reason; nothing changes	Prototype, System	
RUL-04	**Add, enable, disable**	Add a rule; switch one off	The new rule fires; the disabled one does not	Prototype, System	
RUL-05	**Schedule**	Give a rule a date range	Active only between the dates	System	
RUL-06	**Version and audit**	Edit twice	Rule-set version goes up each time; who, when and what changed recorded	Prototype, System	
RUL-07	**Only administrators**	Sign in as Operator	Rules are read-only	System	

7. Test cases — historical query and CSV export

History (HIS)

ID	Test	Steps	Expected result	Where	Result
HIS-01	**Query by location, parameter and dates**	Choose Marrickville and Belmore triangle, rainfall and water level, a 7-day range; **Run query**	Matching records only, newest first, with a total count	Prototype, System	
HIS-02	**Averaging interval**	Run with 1-minute, 10-minute and hourly intervals	Row count and values change to match the interval; values are averages or totals as the parameter needs	System	
HIS-03	**Paging and sorting**	Change rows per page; sort by each column	Pages and order correct across the whole result, not just the page shown	Prototype, System	
HIS-04	**Per-reading status**	Read the Status column	OK, Warning or Alert per reading against the thresholds at that time	Prototype, System	
HIS-05	**CSV export**	**Export CSV**; open in Excel	Exactly the filtered set, with timestamps, units and logger IDs; opens cleanly in Excel	Prototype, System	

8. Test cases — users, roles, sign-in and audit

Users and roles (USR)

ID	Test	Steps	Expected result	Where	Result
USR-01	**Add a user**	**Add user** with name, email, mobile, two roles, two stations	User listed as Invited; the invitation lets them set their own password	Prototype, System	
USR-02	**Edit a user**	Change roles, stations, contact details and notification preferences	Takes effect at the user's next action, without signing out	Prototype, System	
USR-03	**Suspend and reinstate**	Suspend a signed-in user; reinstate	Blocked at once; history kept; reinstated user can sign in again	Prototype, System	
USR-04	**Remove**	Remove a user	Cannot sign in; their past actions still show their name in the audit trail	Prototype, System	
USR-05	**Station scope**	Sign in as a user limited to Belmore triangle	Sees only permitted stations' data and alerts	System	
USR-06	**Six standard roles**	Sign in once per role	Administrator, Operator, Pump Controller, Maintainer, Analyst, Viewer each see and can do exactly what §8.4 lists	System	
USR-07	**Several roles**	Give one user Operator and Pump Controller	They get the union of both, nothing more	System	
USR-08	**Custom role**	Create a role with chosen permissions; assign it	The user gets exactly those permissions	System	
USR-09	**Password policy**	Set a short or common password	Refused with the rule it breaks	System	
USR-10	**Lockout**	Enter a wrong password repeatedly	Account locks after the set number of tries	System	
USR-11	**Inactivity timeout**	Leave a session idle	Warned, then signed out after the set time	System	
USR-12	**MFA (optional)**	Turn MFA on; sign in	A second factor is required	System	
USR-13	**Permission re-check**	Remove a role while its user has a page open; try the action	Refused by the server, not just hidden in the page	System	
USR-14	**Encrypted sessions**	Open the portal over plain http; inspect the session	Redirected to https; the session cookie is secure and cannot be read by page scripts	System	

Audit trail (AUD)

ID	Test	Steps	Expected result	Where	Result
AUD-01	**Everything recorded**	Add a user, change a role, edit a threshold, acknowledge an alert, edit a recipient group	Each recorded with user, time and detail	System	

ID	Test	Steps	Expected result	Where	Result
AUD-02	**Query**	Filter the audit trail by user, action and date	Only matching entries	System	
AUD-03	**Export, no editing**	Export to CSV; try to change an entry	The export matches the filter; entries cannot be edited or deleted	System	

Recipient groups (RCP)

ID	Test	Steps	Expected result	Where	Result
RCP-01	**Add a group**	**Admin** → **Recipients** → **Add group**; save with no name, then with a name, members and channels	Refused until it has a name, a member and a channel; then listed with its members and channels	Prototype, System	
RCP-02	**Edit a group**	Change its members and channels	The list shows the new members and channels; recorded in the audit trail	Prototype, System	
RCP-03	**Remove a group in use**	Remove a group that alert rules send to	Warned which rules use it before removal; after it, those rules no longer send to it	Prototype, System	
RCP-04	**Rules use the groups**	Open a rule's **Recipients & channels**	Shows the current groups, including new and renamed ones	Prototype, System	

9. Test cases — system health, data and fallback

System health and power (HLT)

ID	Test	Steps	Expected result	Where	Result
HLT-01	**Health screen**	Open **Health**	Loggers reporting, conditions raised, lowest battery, pump data gaps; a row per monitored condition	Prototype, System	
HLT-02	**Refresh ≤ 10 min**	Note the last-refresh time over 30 min	Refreshes at least every 10 minutes by itself	System	
HLT-03	**Each fault condition**	Simulate: logger hardware fault, no telemetry for 15 min, one sensor silent, pump data missing	Inoperable, unresponsive, missing sensor data and missing pump data each raised, with a system-fault alert	System	
HLT-04	**Clears by itself**	Restore each fault	Condition clears and the event log records it	System	
HLT-05	**Power telemetry**	Read the station's power panel	State of charge, PV input, charge/discharge and fault flags shown live	System, Site	
HLT-06	**Low-battery leading indicator**	Drop state of charge below the set level	Low-battery alert raised before the station goes offline	System, Site	
HLT-07	**Maintenance mode**	As Maintainer, put a station in maintenance mode, then end it	Everyone sees the station is in maintenance and why; its maintenance alerts can be acknowledged by the Maintainer; recorded in the audit trail	System	

Data acquisition and processing (ACQ)

ID	Test	Steps	Expected result	Where	Result
ACQ-01	**Routine telemetry**	Run the simulator for an hour	Every message stored once, server-timestamped, at the configured cycle	System	
ACQ-02	**Event-triggered messages**	Cross a threshold mid-cycle	Message arrives at once, not at the next cycle	System	
ACQ-03	**Store-and-forward**	Cut the link for 2 h, then restore	The backlog arrives in order; no gaps, no duplicates; alerts in the gap still recorded	System, Site	
ACQ-04	**Range and rate limits**	Send an impossible value and a sudden jump	Flagged or quarantined, not shown as real; data-quality alert raised	System	
ACQ-05	**Frozen sensor**	Send the same value for a long period	Detected as frozen; data-quality alert	System	
ACQ-06	**Radar against float**	Float wet while radar reads low	Sensor-discrepancy alarm	System, Site	
ACQ-07	**Aggregates**	Compare totals with a hand calculation from raw tips	10-min, hourly and 1/3/6/24-hour and 3-day rainfall totals match	System	

Redundancy and fallback (FBK)

ID	Test	Steps	Expected result	Where	Result
FBK-01	**Fault alert first**	Fail the Marrickville rain gauge	A fault alert says alerting has switched to the fallback source	System	
FBK-02	**Rain and wind fallback**	With the gauge failed, rain at Belmore triangle	Alert comes from Belmore, worded to say which sections it now covers	System	
FBK-03	**Temperature fallback**	Fail temperature at a station	Falls back to Belmore triangle, then Bankstown airport and Richmond (BOM)	System	
FBK-04	**Tunnel cross-reading and flood fallback**	Fail one Lady Game Drive radar; fail a flood sensor	Up/down tunnel cross-read; flood falls back to CIDS PTZ monitoring	System	

10. Test cases — responsive layout and system qualities

Phone, tablet and laptop (RSP)

ID	Test	Steps	Expected result	Where	Result
RSP-01	**Columns reflow**	Open Map at phone (~360 px), tablet (~800 px) and laptop widths	One column, two columns, three to four columns; no sideways page scroll	Prototype, System	
RSP-02	**Pump screen on a phone**	Open Marrickville on a phone	One scrollable column in Figure 13's order: readings, health, level trend, pumps, condition, mode, START/STOP	Prototype, System	

ID	Test	Steps	Expected result	Where	Result
RSP-03	**Touch use**	Use menus, filters, pump controls by touch	Every control reachable and large enough to tap; wide tables and charts scroll inside their card	Prototype, System	
RSP-04	**Browsers**	Repeat key journeys in Chrome, Edge, Safari (iPhone) and Chrome (Android)	Same behaviour in each	Prototype, System	

Performance, availability and data (NFR)

ID	Test	Steps	Expected result	Where	Result
NFR-01	**20 concurrent users**	20 named users signed in and active at once	Pages respond normally; alerts still within 5 min	System	
NFR-02	**Hosting and failover**	Review the cloud set-up; restart one server instance	Runs on AWS or Google cloud across redundant infrastructure; service continues	System	
NFR-03	**Backups**	Restore last night's backup to a test copy	Restores completely; readings and settings intact	System	
NFR-04	**Maintenance notice**	Schedule maintenance	Users are told at least 48 h ahead	System	
NFR-05	**Data kept permanently**	Query the oldest stored data	Still present and returned by the query engine; nothing purged	System	

Availability over time (not inoperable more than 12 h in any 6 months, ≤ 24 h cumulative per 12 months, one 36 h maintenance window a year; §9) is measured in service from the health records, not in a test session.

11. Sign-off

The plan has 113 tests across 16 areas. A release is accepted when every test has run and no Critical or Major defect is open.

Before testing starts

- MTS has confirmed the thresholds, alert wording, recipients and alert channels marked *(pending)*.
- The server is built and loaded with the station simulator; test accounts exist for each of the six roles.

Testing is complete when

- Every test is marked Pass, or Fail with a logged defect.
- No Critical or Major defect is open; Minor defects have an agreed fix date.
- Site tests (PMP-08, HLT-05 and 06, ACQ-03 and 06) have passed at each location during the site acceptance test.

Role	Name	Date	Signature
Veldora			
Metro Trains Sydney			